INSIGHT | ACTION | CONTROL

ISPE UK · TAKE-AWAY CHECKLIST

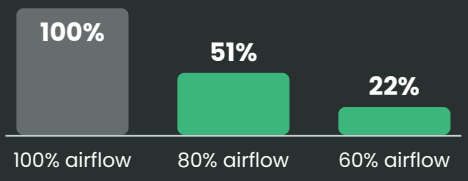
2-MINUTE SELF-CHECK

Is your cleanroom wasting energy?

Tick every statement that's true of your site. Each tick points to a saving you can usually make **while staying in grade**.

WHY AIRFLOW MATTERS

-20% airflow
≈ -49% fan power



Fan power ≈ airflow³ (fan affinity laws, ideal case). Small airflow cuts, big energy savings.

Operational efficiency

- [ ] Air change rates haven't been reviewed since commissioning.
- [ ] Fans run at full speed overnight and at weekends.

Sustainability & carbon

- [ ] We don't know what share of site energy our cleanrooms use.
- [ ] Our carbon targets leave GMP areas out of scope.

Digitalisation & data

- [ ] Particle data is logged for compliance, never used to control airflow.
- [ ] We can't see energy use per room or suite.

Compliance

- [ ] Energy projects stall because nobody's sure how QA will react.
- [ ] Energy measures aren't part of our Contamination Control Strategy.

Future-ready design

- [ ] Our next build or refit is designed around fixed, worst-case airflow.
- [ ] Our BMS can't adjust airflow to live room conditions.

COUNT YOUR TICKS

0–2
Well tuned
Prove it: benchmark your site against your peers.

3–5
Quick wins likely
Start with the three checks overleaf.

6–10
Big opportunity
Worth a conversation. Scan the QR code overleaf.

Turn over: 3 checks for when you're back on site →

Guidance only, not a site assessment.

THREE CHECKS FOR NEXT WEEK

Back on site? Start here.

1

Check the headroom

Pull a month of particle monitoring data. If counts sit well below grade limits, at rest and in operation, you're likely moving more air than you need.

2

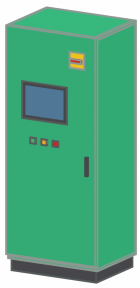
Check the night shift

Do fans step down when rooms are empty? A validated setback mode is often the fastest, lowest-risk win.

3

Check in with QA

Assess each measure against your Contamination Control Strategy, like any other GMP change. Savings and compliance get signed off together.



HOW EECO2 HELPS

Integrated Cleanroom Control Systems (ICCS®)

ICCS matches airflow to real-time particle counts, cutting fan energy while your rooms stay in grade and GMP compliant.

INSIGHT

We analyse your particle, HVAC and energy data to find surplus airflow.

ACTION

We make each change with your QA team, under change control.

CONTROL

Continuous monitoring keeps savings inside your validated limits.

[RESULT: e.g. "Grade C suite: X% less fan energy, stayed in grade, change-controlled in Y weeks."]

How many did you tick? Let's talk them through.

Scan to tell us about your facility. One of our engineers will get back to you to discuss where the savings are, and how to make them without putting compliance at risk.

SCAN TO TALK TO US

eeco2.com/contact

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References: ISO 14644-16:2019 Cleanrooms and associated controlled environments – Energy efficiency; EU GMP Annex 1 (2022) Manufacture of Sterile Medicinal Products; fan affinity laws (ideal case). Guidance only, not a site assessment.